



Seismic Interferometry

Chapter 10

§10.1 Introduction

To understand the Earth's internal structure, the propagation of seismic waves is a crucial clue. Traditionally, seismic velocity structures have been imaged by observing the ground motions caused by earthquakes. Since the 2000s, it has become common to investigate the Earth's interior by analyzing random ground motions induced by phenomena other than earthquakes, a technique known as Seismic Interferometry ^{note 1}. In this chapter, based on the knowledge we have learned so far, we will explain the principles of seismic interferometry.

In this text, we will focus on microseisms induced by ocean swells as the random wavefield (see Section 4.7.3). Another important random wavefield is the seismic coda wave (multiply scattered seismic waves; for details, see Sato, Fehler and Maeda, 2012⁽³⁰⁾). Seismic interferometry using coda waves is also widely researched (e.g., Campillo and Paul, 2003⁽⁵⁾), but we will omit the details here due to time constraints. If you are interested, please refer to the aforementioned review papers and textbooks.

^{note 1}) See recent review papers (e.g., Snieder and Larose, 2013⁽³⁶⁾) and textbooks (e.g., Schuster, 2009;⁽³¹⁾ Sato, Fehler and Maeda, 2012;⁽³⁰⁾ Nakata et al., 2019⁽²¹⁾). Seismic interferometry is not limited to the Earth but has been applied to various phenomena, including laboratory scales (e.g., Lobkis and Weaver, 2001⁽¹⁷⁾), architectural structures (e.g., Snieder and Wapenaar, 2010⁽³⁴⁾), helioseismology (e.g., Gizon et al., 2010⁽¹¹⁾), and ocean acoustics (e.g., Roux and Kuperman, 2004⁽²⁹⁾).

§10.2 A brief history of Seismic Interferometry

The idea of seismic interferometry dates back to the 1950s. The spatial autocorrelation (SPAC) method proposed by Aki (1957)⁽²⁾ pioneered the concept of seismic interferometry. There are also pioneering works in the fields of ocean acoustics by Cox (1973)⁽⁷⁾ and exploration seismology by Claerbout (1968).⁽⁶⁾ Aki's idea, however, remained unnoticed for a while^{note 2)}. However, 26 years after its publication, in 1983, a research group led by Hiroshi Okada from Hokkaido University paid attention to Aki's method (Okada and Sakajiri, 1983), and it became actively studied primarily as a technique (microtremor survey) to investigate ground characteristics near the surface. Since then, the microtremor survey has become a standard method for estimating very shallow structures near the surface.

The paper that triggered the attention toward seismic interferometry in the field of seismology was Campillo and Paul [2003].⁽⁵⁾ They analyzed seismic waves generated in Mexico and demonstrated that the propagation of surface waves could be extracted by computing the cross-correlation of the well-scattered coda wave portion of the seismic records. In 2005, Shapiro et al.⁽³³⁾ took advantage of the fact that microseisms continuously arrive from various directions, and succeeded in estimating the crustal structure of California from how these waves propagate. This technique is called ambient noise tomography. Following this study, similar research has been conducted in numerous regions, including North America, Japan, China, and Europe.

The fundamental observable in seismic interferometry is the cross-correlation function of seismic waveforms between two selected stations. The waveform of the cross-correlation function can be interpreted as if there is a virtual source at one station, and the waveform is recorded at the other station. Let us look at Figure 10.1. It shows the cross-correlation functions between the station in Matsushiro and others. This method has two major advantages.

The first advantage is the ability to virtually place an event even in regions without earthquakes. Usually, in aseismic regions, detailed seismic velocity structures are unknown, so the fact that earthquakes are not required is highly beneficial. Figure 10.1 indeed shows the propagation of Rayleigh waves as if Matsushiro were the epicenter.

The second advantage is that there is no need to wait for an earthquake. Normally, to perform seismic wave tomography analysis, one must wait for a sufficient amount of earthquake data to accumulate. In seismic interferometry, a sufficient quality of data can be secured by observing for a certain period^{note 3)}. This property, not needing to wait for an earthquake, is extremely advan-

note 2) Aki's work was compiled as his doctoral dissertation. According to his autobiography, the idea was inspired by Wiener's Cybernetics. For his thesis, he independently handled the theory, the design and construction of observational equipment, data collection, and structural estimation. Because this was before the widespread use of computers, he even had to design analog circuits to calculate cross-correlation coefficients. He conducted the observations at the athletic field of the Faculty of Science, The University of Tokyo, and ultimately dug into the field himself to verify the structural estimation results.

note 3) Depending on the frequency band, roughly speaking, several months is sufficient for 0.05-0.5 Hz, and a few days is

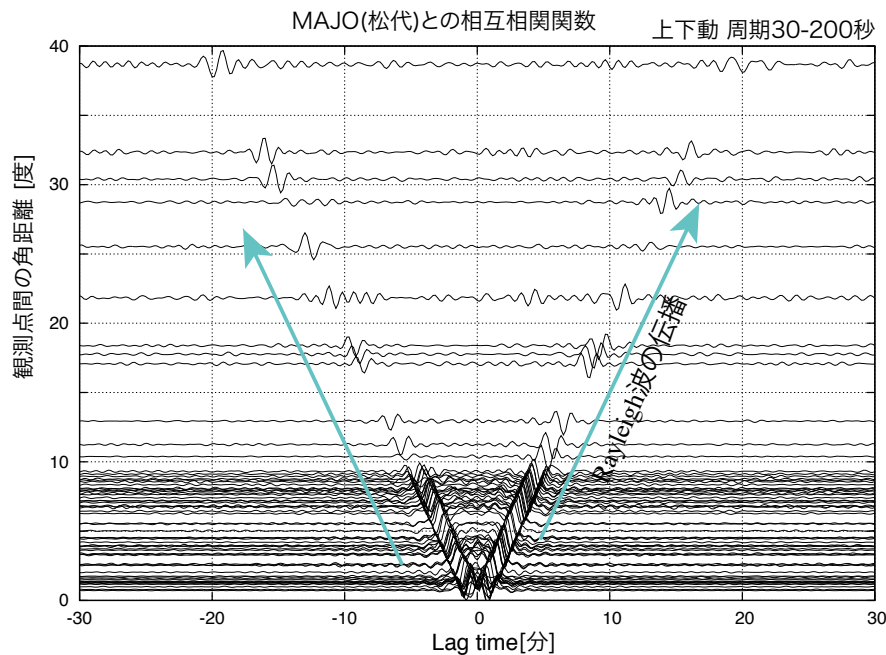


Figure 10.1: Cross-correlation functions of vertical-component records between MAJO (Matsushiro) and other stations, arranged by inter-station distance. A band-pass filter of 20-200 seconds is applied. The propagation of Rayleigh waves is clearly visible. The wave packet with a positive delay time is called the causal part, and the wave packet with a negative delay time is called the acausal part.

tageous for investigating temporal changes in seismic velocity structures. To detect infinitesimal temporal changes in velocity structure using earthquakes, one must wait for earthquakes to occur repeatedly at the exact same location (known as repeating earthquakes). However, such convenient earthquakes occur very rarely. Calculating the cross-correlation function of a certain pair of stations and observing its temporal variation is equivalent to analyzing records of earthquakes repeatedly occurring at the same location. In fact, temporal structural changes accompanying volcanoes and earthquakes are now actively researched (e.g., Sens-Schönfelder and Wegler, 2006⁽³²⁾).

Here, we will explain the theoretical background and the principles of seismic interferometry. For specific application examples, please refer to the review papers mentioned at the beginning.

Rough Earth Club and Smooth Earth Club

SI brings a reunion of different research fields: ocean acoustic (Cox, 1973), seismic exploration (Claebout, 1968), and seismology (Aki 1957). Surprisingly the ideas were proposed independently and simultaneously.

often enough for higher frequencies. Of course, a longer observation period is more advantageous to ensure sufficient accuracy.

Even in the seismological community, there were different cultures: one is the rough Earth club and smooth Earth club. Keiiti Aki defined them in his letter to V. I. Keilis-Borok, as

... To a geodynamicist, the earth's property is smoothly varying within bodies bounded by large-scale interfaces. Most seismologists also belong to this "smooth earth club" because once you start with an initial model of smooth earth, your data usually do not require the addition of small-scale heterogeneity to your initial model. As summarized well in a recent book by Sato and Fehler (1998), the acceptance of coda waves in the data set is needed for the acceptance of small-scale seismic heterogeneity of the lithosphere. There is an increasing number of seismologists who accept it, forming the "rough earth club." I believe that you are also a member of the rough earth club, judging from the emphasis on the hierarchical heterogeneity of the lithosphere. . .

("Seismology of Earthquake and Volcanic Prediction", Lecture notes, Aki 2003). Seismic wavefield above 1 Hz was a territory of the "rough Earth club," whereas that below 0.1 Hz was a territory of the "smooth Earth club." The members of the rough Earth club are familiar with stochastic treatments of the seismic wavefield. The recent development of SI means a reunion between the "rough Earth club" and the "smooth Earth club."

The dominant frequency of microseisms at around 0.2 Hz corresponds to the gap between "*rough Earth club*" and "*smooth Earth club*". SI enables us to utilize coherent signals from random seismic wavefields with an assumption of stochastic stationary excitation. Although surface wave tomography was a tool of the "smooth Earth club," SI broke the gap. Scattering due to strong lateral heterogeneities in the crust and the sediment was a big barrier for "*smooth Earth club*." When we apply SI, the scatterer is important because it enhances the randomness. SI plays a complementary role in the "smooth Earth club." This role of SI is true of other communities, such as seismic exploration, acoustic, physical oceanography, and so on.

In this lecture, the next chapter explains the excitation mechanism of microseisms by ocean swell. The next chapter explains the basic principle of SI with a demonstration by a WEB application. Then the last chapter explains some applications, which are done by our group mainly.

§10.3 Theoretical background of Seismic Interferometry: a closed system

To understand the theory of seismic interferometry, let us consider a simple case ^{note 4)}.

First, let us review the definition of the cross-correlation function. I have created a web application (<http://www.eri.u-tokyo.ac.jp/knishida/Seismology/wave2Drandom2.html>), and it would be easier to understand if you read along while interacting with it interactively.

10.3.1 Cross-correlation analysis

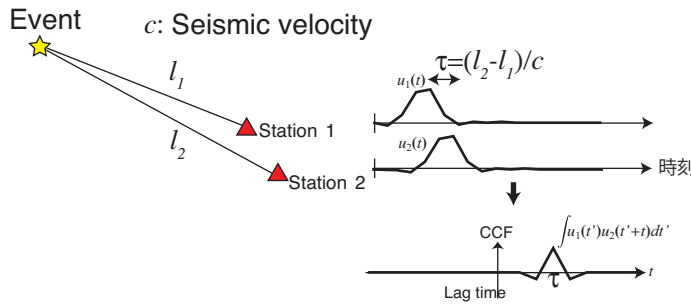


Figure 10.2: Schematic figure of cross-correlation analysis for a pair of seismograms when an earthquake occurred.

First, we consider a transient phenomenon such as an earthquake or a volcanic explosion. Let us suppose an earthquake occurs at a certain location, as shown in Figure 10.2. For seismograms at stations 1 and 2, we define the cross-correlation function ϕ_{12} as

$$\phi_{12}(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T u_1(\tau) u_2(t + \tau) d\tau. \quad (10.1)$$

Assuming that the cross-correlation function quickly converges to 0 as $|t|$ increases, we can define its Fourier transform $\Phi_{12}(\omega)$ as

$$\Phi_{12}(\omega) = \langle U_1^*(\omega) U_2(\omega) \rangle \quad (10.2)$$

^{note 4)}For a comprehensive understanding of the theoretical treatment, Snieder et al. (2010)⁽³⁷⁾ is highly recommended.

note 5). In this case, the wave arrives at station 2 delayed by τ seconds relative to station 1 (Figure 10.2, right). As a result, the cross-correlation function has a peak at $t = \tau$. In other words, by reading the time of the peak of the cross-correlation function, we can measure the difference in seismic wave arrival times (travel-time difference). In practice, precise travel-time differences measured by cross-correlation functions are often utilized to determine hypocenter locations accurately.

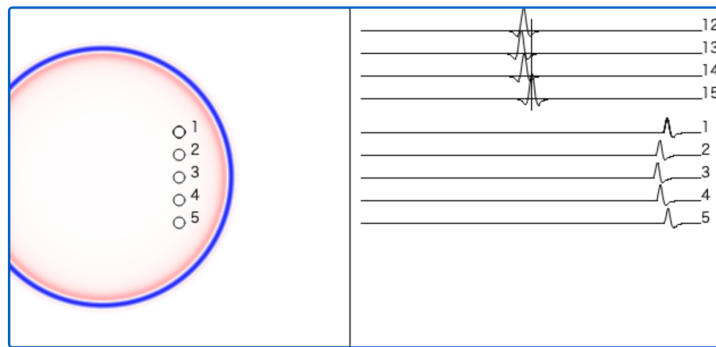


Figure 10.3: Example of simulation results with a point source. The bottom right panel shows the waveforms observed at stations 1 to 5, and the top right panel shows their cross-correlation functions.

In the web application, clicking allows you to place a seismic source at any location, and pressing 's' starts the simulation. In the bottom right, waveforms at stations 1-5 are displayed, and in the top right, their cross-correlation functions (for example, '14' represents the cross-correlation function between 1 and 4) are shown. By reading the peak time of the cross-correlation function, you can measure the travel-time difference. Please give it a try.

Hereafter, we will consider phenomena that can be approximated as statistically random and stationary, such as microtremors and ambient noise, rather than transient phenomena (earthquakes and volcanic eruptions).

10.3.2 SI in a closed system

Here, we consider SI in a closed system. In the case of a finite body, we evaluate CCFs based on a normal mode approach.⁽¹⁷⁾ Because the Earth is a finite-size sphere, this approach is also feasible for multi-orbit propagations on a global scale.^{(24), (25)} For simplicity, this subsection describes a scalar 1-D case, but it can be easily extended to elastic 2-D and 3-D cases.

note 5) Strictly speaking, when considering a stationary process, the long-term integration is approximated by the ensemble average. Since we are assuming that the wavefield is stationarily excited, $u_i(t)$ is not square-integrable, and therefore its Fourier transform cannot be taken in the usual sense. If you wish to understand it more rigorously, please refer to reference books on time series analysis.

Repeating seismic experiments

Virtually, we consider repeating seismic experiments in a closed system of a perfect elastic body. Before the initial time $t = 0$, the body did not deform, and a random force $f^k(x)$ was applied to the body at $t = 0$ in the k th experiment, where x is the spatial location. We consider a finite body from $x = 0$ to $x = L$ and impose rigid or free boundary conditions at both ends. The equation of motions is given by

$$\rho(x) \frac{\partial^2 u^k(x, t)}{\partial t^2} = \kappa(x) \frac{\partial^2 u^k(x, t)}{\partial x^2} + f^k(x) \delta(t), \quad (10.3)$$

where κ is the elastic constant, ρ is the density and u^k is the displacement at the k th experiment. After the force is applied, the displacement is measured at the stations. These experiments are repeated K times. Note that we cannot consider a persistent force in this system because no attenuation leads to an infinite increase in the amplitude over time.

We evaluated the displacement $u^k(x, t)$ by a convolution between the force $f^k(x)$ and the Green's function. The Green's function can be written in terms of normal mode theory⁽⁸⁾ as

$$G(x, x'; t) = \sum_n \frac{\mathcal{U}_n(x) \mathcal{U}_n(x')}{\omega_n} \sin(\omega_n t), t \geq 0, \quad (10.4)$$

where ω_n is n th eigenfrequency and $\mathcal{U}_n$ is n th eigen function, which satisfies orthonormality:

$$\int_0^L \rho(x) \mathcal{U}_n(x) \mathcal{U}_{n'}(x) dx = \delta_{nn'}. \quad (10.5)$$

Then, the displacement can be represented by the convolution

$$u^k(x, t) = \sum_n A_n^k \mathcal{U}_n(x) \frac{\sin(\omega_n t)}{\omega_n}, \quad (10.6)$$

where $A_n^k = \int_0^L \mathcal{U}_n(x) f^k(x) dx$. We consider the energy partition of the modes for a random external force in the next section.

Energy partition of modal energy

This subsection discusses the energy balance of each mode: how the work done by external forces is distributed to the kinematic and elastic energy. First, we evaluate the work done by the external force can be given by

$$\int_0^\infty \int_0^L v_n^k(x, t) f^k(x) \delta(t) dx dt = (A_n^k)^2, \quad (10.7)$$

where the particle velocity v_n^k of n th mode in the k th experiment is written by $v_n^k = A_n^k \mathcal{U}_n(x) \cos(\omega_n t)$.

The kinetic energy of n th mode T_n can be evaluated by integrating the kinetic energy density in space as

$$T_n = \int_0^L \frac{\rho}{2} (v_n^k)^2 dx = \frac{1}{2} (A_n^k)^2 \cos^2(\omega_n t). \quad (10.8)$$

The elastic energy V_n , on the other hand, can be evaluated by integrating the strain energy density in space. The partial integral with the boundary condition leads to

$$V_n = \int_0^L \frac{\kappa}{2} \left(A_n^k \frac{\partial \mathcal{U}_n}{\partial x} \frac{\sin(\omega_n t)}{\omega_n} \right)^2 dx = \frac{1}{2} (A_n^k)^2 \sin^2(\omega_n t). \quad (10.9)$$

The total energy $T_n + V_n$ is

$$T_n + V_n = (A_n^k)^2, \quad (10.10)$$

which balances the work done by external forces.

Here, we consider an excitation by random force F_i^k at location x_i as

$$f^k(x) = \sum_{i=0}^{I-1} F_i^k \delta(x_i), \quad (10.11)$$

where I is the number of the force. We also assume that $F_i^k(x)$ is white noise as

$$\langle F_i^k F_{i'}^k \rangle_k = \bar{F}^2 \delta_{ii'}, \quad (10.12)$$

where $\langle \rangle_k$ is ensemble average with respect to k .

White noise

Here we will consider an external force f applied randomly. Let us define the external force as random fluctuations that contain all wavenumber components equally. Having such property is called white noise^{note 6)}. Therefore, its power spectrum $\Phi(f)$ is constant.

Let us consider this more concretely. Look at the left diagram in Figure 10.4. Consider an external force $f^k(i\Delta x)$ discretized with a spatial interval Δx . At each position $i\Delta x$, we roll a dice randomly (assuming it follows a normal distribution with mean 0 and variance 1). Now let's consider the autocorrelation function $\phi(x, x')$. We define the autocorrelation function as

$$\phi(x, x') = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} f^k(x') f^k(x' + x). \quad (10.13)$$

Now, we assume that there is absolutely no correlation in the external force if the distance between two points is separated by $\Delta x/2$. Then ϕ becomes a function of only $x - x'$, and letting $\xi \equiv x - x'$, we get

$$\phi(\xi) = \begin{cases} 1 & |\xi| \leq \Delta x \\ 0 & \text{otherwise.} \end{cases} \quad (10.14)$$

^{note 6)} This is an analogy with sunlight. Generally, white implies containing all frequency components, red implies being rich in low frequencies, and blue implies being rich in high frequencies.

By expanding $\phi(\xi)$ into a Fourier series, we obtain $\Phi(k_n) = \Delta x$, where k_n is the wavenumber $2n\pi/L$. The power spectrum (one-sided spectrum) is $2\Delta x$ (Figure 10.4).

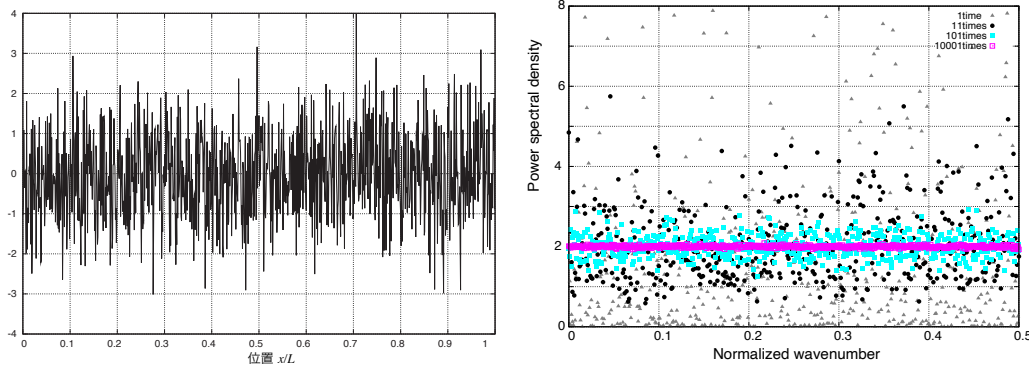


Figure 10.4: Left: external force $f^k(x)$ with white noise characteristics. Right: the result of calculating the power spectrum. As the number of ensemble averages increases, the estimated value approaches 1.

The theoretical calculation has been done. What happens if we actually create numerical data and calculate it? Let's generate white noise using random numbers and perform Fourier analysis.

First, let's consider one extracted window. Let's consider the Fourier spectrum at wavenumber k_n ^{note 7)}. Since $f(x_k)$ is a random number, its Fourier component $F(k)$ is also a random number. Since the power spectrum can be calculated as $|F(k)|^2$, it does not take a constant value but takes a random value ^{note 8)}. The gray dots on the right of Figure 10.4 are the actually calculated points. This is slightly strange. It is supposed to be $\Phi(f) = 1$, but with just one discrete Fourier transform, we can only see that it is on the same order of magnitude as 2. Why is this?

This is because the autocorrelation function is a statistical quantity. It does not become a meaningful value unless averaged over many ensembles (here k is the index assigned to ensemble elements). More specifically, the estimation error becomes 100%. Although it seems very odd at first glance, this behavior arises because we are statistically evaluating only one sample.

So let us perform the Fourier series expansion many times and take the ensemble average. You will see that it gradually approaches 2 ^{note 9)}. As such, the power spectrum is ultimately a statistical quantity, so the estimation error becomes too large unless averaged sufficiently. For example, consider the case where we sample and average N times. If the time series follows a

^{note 7)}For actual calculation, space must be discretized and a discrete Fourier transform must be performed. Ensuring consistency between the discrete Fourier transform and the Fourier series expansion gives

$$F(k_n) = \Delta x \sum_{j=0}^{N-1} f(x_j) e^{-i2\pi k_n x_j} = \Delta x F^{DFT}(k_n).$$

^{note 8)}More precisely, it follows a χ^2 distribution.

^{note 9)}Just to confirm, we are currently considering a time series with a variance of 1 in the time domain. From Percival's formula, the integral of the power spectrum becomes 2×0.5 , which matches.

normal distribution, the estimation error is $1/\sqrt{N}$. The way the error decreases according to N can also be seen from the figure.

Equipartition of energy

To simplify the problem, we consider the constant density ρ_0 . The expected value of the cross-correlation of A_n can be evaluated as follows.

$$\langle A_n^k A_{n'}^k \rangle_k \sim \frac{I\bar{F}^2}{L\rho_0} \int_0^L \mathcal{U}_n(x) \mathcal{U}_{n'}(x) dx = \frac{I\bar{F}^2}{L\rho_0} \delta_{nn'} \equiv \mathcal{E} \delta_{nn'}, \quad (10.15)$$

where $\mathcal{E}$ is the modal energy. The amplitudes of the different modes A_n do not correlate with each other, and the total energy of each mode is distributed equally. The expected value of the modal energy is constant for each mode when the external force f is white noise. When a system meets this condition, we call the state the equipartition of energy. We note that the fluctuation-dissipation theorem in physics⁽⁴⁾ requires a thermal equilibrium that satisfies the equipartition of energy. Equation 10.12 can be derived from the principle of equal a priori probabilities in the case.

CCFs under the equipartition of energy

Here we define a CCF $\phi^k(x_1, x_2; \tau)$ between $u^k(x_1)$ and $u^k(x_2)$ under the equipartition of energy as,

$$\phi^k(x_1, x_2; \tau) \equiv \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T u^k(x_1, t) u^k(x_2, t + \tau) dt, \quad (10.16)$$

where x_1 shows the location of station 1, and x_2 shows that of station 2. We note that there are two types of CCF definitions. The sign of the second term of the other type is flipped.

The ensemble average of $\phi(x_1, x_2; t)$ over K time experiments is defined by the ensemble average $\langle \phi^k(x_1, x_2; \tau) \rangle_k$. Then, the CCF can be represented by

$$\frac{d}{d\tau} \phi(x_1, x_2; \tau) = -\frac{\mathcal{E}}{2} (G(x_1, x_2; \tau) - G(x_2, x_1; -\tau)), \quad (10.17)$$

which relates the CCF to Green's function.⁽³⁵⁾ The frequency dependence of this equation differs slightly from Aki's formulation (e.g.,⁽¹³⁾). Although the formulation showed that the Hilbert transform of the CCF can be related to Green's function, the difference can be attributed to spectral normalization.

Compared to CCFs for real data, the biggest problem is the assumption of the equipartition of energy. Because the excitation sources are distributed near the surface, fundamental modes dominate the observed wave field. Energy is not partitioned equally among different mode branches. Indeed, the observed dominance of fundamental modes (e.g.,⁽²²⁾) shows that the energy is not equally distributed in the radial direction. When considering a single-mode branch, the energy is distributed equally in the horizontal direction if the external force is white noise distributed on the entire surface.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

Although the theory assumed K experiments, practically, only one observation is possible. Therefore, the ensemble averages need to be replaced by time averages to calculate the CCFs of observed data for persistent external forces (e.g., ocean waves). However, no attenuation in a closed system causes the problem of diverging amplitudes without attenuation; that is, the seismic wave field in a closed system never meets the equilibrium for persistent sources. Physically, it is natural to consider the equilibrium between the energy dissipation due to attenuation and the work performed by external forces.^{(10), (14)}

We note that the term Green's function is often used in a less mathematically rigorous sense in studies on SI. Mathematically, a CCF converges to the Green's function only in limited cases. A similar situation occurs in quantum field theory. The Green's function refers to CCFs, although it does not satisfy the mathematical definition.⁽⁵⁰⁾

10.3.3 CCFs for homogeneous source distribution in a 2-D homogeneous medium

For ANT, we explicitly express the CCFs between all pairs of three-component seismometers in an open system when multimode Love and Rayleigh waves dominate the ambient seismic wave field. We evaluated the CCFs for the surface waves in a 2-D problem with a membrane approximation (e.g.,^{(39), (41)–(43)}). We show an expression of the mixed-component CCFs of surface waves.

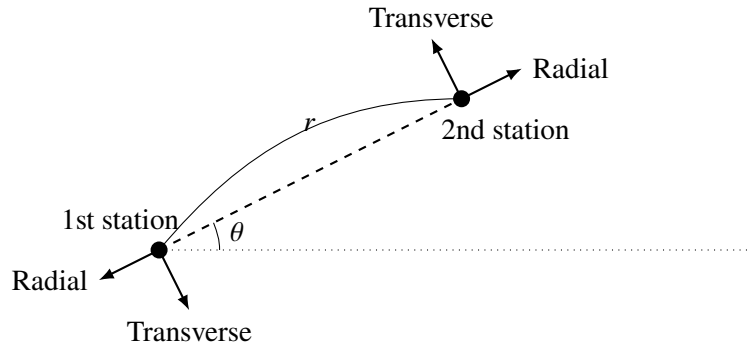


Figure 10.5: Schematic of the geometry of a pair of stations with separation distance r and the coordinate. The figure shows the radial and transverse directions for the station pair. To simplify the equation, the signs of the directions are different from those of other studies. The difference changes the signs of CCF in the components of RR, TT, RZ, and ZR.

An arbitrary seismic wave field $\mathbf{u}(r, \theta; \omega)$ in 2-D can be represented by a superposition of multimode Love and Rayleigh waves as

$$\mathbf{u}(r, \theta; \omega) = \sum_{n=0}^{\infty} \sum_{m=-\infty}^{\infty} f_{n,m}^{Ray}(\omega) [U_n(\omega) \mathbf{P}_m(r, \theta; \omega) + V_n(\omega) \mathbf{B}_m(r, \theta; \omega)] + f_{n,m}^{Love}(\omega) W_n(\omega) \mathbf{C}_m(r, \theta; \omega), \quad (10.18)$$

where $f_{n,m}^{Ray}(\omega)$ is forcing for Rayleigh waves, and $f_{n,m}^{Love}$ is forcing of Love waves. U_n and V_n are eigenfunctions of the n th overtone of the Rayleigh wave, and W_n is the eigenfunction of the n th overtone of the Love wave, which has a real value. The basis functions $\mathbf{P}_m$, $\mathbf{B}_m$ and $\mathbf{W}_m$ are given by

$$\mathbf{P}_m(r, \theta; \omega) = \hat{\mathbf{z}} J_m \left(\frac{\omega r}{c_n^{Ray}(\omega)} \right) e^{im\theta}, \quad (10.19)$$

$$\mathbf{B}_m(r, \theta; \omega) = \frac{c_n^{Ray}(\omega)}{\omega r} \left[\hat{\mathbf{r}} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} \right] J_m \left(\frac{\omega r}{c_n^{Ray}(\omega)} \right) e^{im\theta}, \quad (10.20)$$

$$\mathbf{C}_m(r, \theta; \omega) = \frac{c_n^{Love}(\omega)}{\omega r} \left[\hat{\mathbf{r}} \frac{\partial}{\partial \theta} - \hat{\theta} \frac{\partial}{\partial r} \right] J_m \left(\frac{\omega r}{c_n^{Love}(\omega)} \right) e^{im\theta}, \quad (10.21)$$

where $c_n^{Ray}(\omega)$ phase velocity of n th overtone of Rayleigh wave, and $c_n^{Love}(\omega)$ is n th overtone of Love wave.

An assumption of equipartition of energy leads to

$$\langle f_{n,m}^{Ray*} f_{n',m'}^{Ray} \rangle U_n(\omega) U_{n'}(\omega) \equiv P_n^{Ray}(\omega) \delta_{nn'} \delta_{mm'}, \quad (10.22)$$

$$\langle f_{n,m}^{Love*} f_{n',m'}^{Love} \rangle W_n(\omega) W_{n'}(\omega) \equiv P_n^{Love}(\omega) \delta_{nn'} \delta_{mm'}, \quad (10.23)$$

$$\langle f_{n,m}^{Ray*} f_{n',m'}^{Love} \rangle = \langle f_{n,m}^{Love*} f_{n',m'}^{Ray} \rangle = 0, \quad (10.24)$$

$$R_n(\omega) \equiv V_n(\omega) / U_n(\omega), \quad (10.25)$$

where $P_n^{Ray}(\omega)$ is the power spectrum of n th overtone of Rayleigh wave, and P_n^{Love} is the power spectrum of n th overtone of Love wave.

The mixed-component cross-spectra for multimode Rayleigh waves are written as,

$$\begin{bmatrix} \Phi_{ZZ}^{Ray} & \Phi_{ZR}^{Ray} & \Phi_{ZT}^{Ray} \\ \Phi_{RZ}^{Ray} & \Phi_{RR}^{Ray} & \Phi_{RT}^{Ray} \\ \Phi_{TZ}^{Ray} & \Phi_{TR}^{Ray} & \Phi_{TT}^{Ray} \end{bmatrix} = \sum_{n=0}^{\infty} \left\{ P_n^{Ray}(\omega) \begin{bmatrix} J_0 \left(\frac{\omega r}{c_n^{Ray}} \right) & -R_n J_1 \left(\frac{|\omega| r}{c_n^{Ray}} \right) & 0 \\ -R_n J_1 \left(\frac{|\omega| r}{c_n^{Ray}} \right) & -\frac{R_n^2}{2} J_{0-2} \left(\frac{\omega r}{c_n^{Ray}} \right) & 0 \\ 0 & 0 & -\frac{R_n^2}{2} J_{0+2} \left(\frac{\omega r}{c_n^{Ray}} \right) \end{bmatrix} \right\}, \quad (10.26)$$

where a cross-spectrum represents the CCF in frequency domain. The mixed-component cross-spectra for multimode Love waves are written as,

$$\begin{bmatrix} \Phi_{ZZ}^{Love} & \Phi_{ZR}^{Love} & \Phi_{ZT}^{Love} \\ \Phi_{RZ}^{Love} & \Phi_{RR}^{Love} & \Phi_{RT}^{Love} \\ \Phi_{TZ}^{Love} & \Phi_{TR}^{Love} & \Phi_{TT}^{Love} \end{bmatrix} = \sum_{n=0}^{\infty} \left\{ P_n^{Love}(\omega) \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{1}{2}J_{0+2}\left(\frac{\omega r}{c_n^{Love}}\right) & 0 \\ 0 & 0 & -\frac{1}{2}J_{0-2}\left(\frac{\omega r}{c_n^{Love}}\right) \end{bmatrix} \right\}, \quad (10.27)$$

where $J_{0-2}(z) \equiv J_0(z) - J_2(z)$ and $J_{0+2}(z) \equiv J_0(z) + J_2(z)$. Z represents the vertical component, and R and T represent the horizontal components according to the polarization direction (Figure 10.5).

These equations are identical to the result of.⁽¹³⁾ A similar formulation for DAS observation is given by.⁽²⁰⁾ We note that the difference of sign Φ_{RZ}^{Ray} , Φ_{RR}^{Ray} and Φ_{TT}^{Ray} originates from the definition of the radial and transverse directions, which differ from this study. Using this formulation, cross-spectra for Rayleigh waves lead to a symmetric relation between the RZ and ZR components as $\Phi_{RZ}^{Ray} = \Phi_{ZR}^{Ray}$, whereas cross-spectra for body waves lead to an antisymmetric relation as $\Phi_{RZ}^{body} = -\Phi_{ZR}^{body}$.⁽³⁸⁾

Attenuation becomes significant when considering a seismic wave field in sediment above 0.1 Hz.^{(23), (27), (28)} However, physically plausible attenuation measurements are practically difficult (e.g.,^{(15), (45)}) because there is ambiguity between the source heterogeneities and the attenuation. Although new techniques have been developed to overcome this problem (e.g.,^{(3), (16), (18)}), they are beyond the scope of this study. Even if the physically plausible attenuation estimation is difficult, apparent Q measurements are feasible for better fitting a synthetic cross-spectrum to an observation (e.g.,⁽²³⁾).

§10.4 SI in an open system

Next, we consider SI in an open system. The formulations in an open system depend on the source-receiver configuration: (1) random point sources distributed over the whole space, (2) random sources distributed on a closed curve, and (3) uncorrelated plane wave incidents from various directions. This subsection describes the relationship between the different configurations of an open system. Mathematically, the theory of an open system differs from that of a closed system. We cannot use normal mode theory for an open system because the system loses energy from the radiation boundary.

For ANT, CCFs are usually formulated in an open system, which better approximates the source-receiver configuration. Because the surface wave can be formulated as a 2-D problem with a membrane approximation (e.g.,^{(39), (41)–(43)}), we can consider them as a 2-D potential problem in an open system. For ANT, we explicitly express the mixed-component CCFs of surface waves in a simplified case at the end of the subsection.

10.4.1 Seismic excitation by an infinite number of sources

We consider one realization of the background seismic wave field excited by an infinite number of sources in an open system. When external force acts on the body at a location $\mathbf{r}$ and time $t = 0$, the 2-D wave equation is given by

$$\frac{\partial^2 \psi(\mathbf{r}, t)}{\partial t^2} = C(\mathbf{r}) \nabla^2 \psi(\mathbf{r}, t) + \sum_{i=0}^{\infty} \delta(\mathbf{r}_i - \mathbf{r}) \delta(t) f_i \quad (10.28)$$

where $C(\mathbf{r})$ is phase velocity, and f_i is i th external force at the location $\mathbf{r}_i$ ($i = 0, \dots, \infty$). The potential $\psi(x, t)$ is given by

$$\psi(\mathbf{r}, t) = \sum_{i=0}^{\infty} g^{2D}(\mathbf{r} - \mathbf{r}_i; t) f_i, \quad (10.29)$$

where g^{2D} is the Green's function in time domain.

The Fourier component of the potential $\psi(\mathbf{r}, t)$ is written by

$$\Psi(\mathbf{r}, \omega) = \sum_{i=0}^{\infty} G^{2D}(\mathbf{r} - \mathbf{r}_i; \omega) f_i, \quad (10.30)$$

where G^{2D} is the Green's function in frequency domain. In this paper, we define the Fourier transform as follows,⁽⁸⁾

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt. \quad (10.31)$$

We note that the Fourier convention depends on the literature. For example,⁽¹⁾'s definition has the opposite sign in the exponential term.

Here, we consider a potential $\psi(\mathbf{r}_o, t_0)$ in a simplified case with constant phase velocity as $C(\mathbf{r}) = C_0$, where $\mathbf{r}_o$ is the location of the origin and time denotes an arbitrary positive time. The $\psi(\mathbf{r}_o, t_0)$ is represented by the sum of the arrivals excited by sources along the concentric circle with radius $r = C_0 t_0$. The typical separation of the sources is assumed to be Δx . Within the circle with the band Δx , about $2\pi r / \Delta x$ sources are distributed (Figure 10.6 left). Because the amplitude decay is proportional to $r^{1/2}$, the mean square amplitude $\psi(\mathbf{r}_o, t_0)$ is estimated to be about $2\pi r / (\Delta x (r^{1/2})^2) = 2\pi / \Delta x$, which does not depend on the distance r . Therefore, after $t > 0$, the fluctuations of ψ last for a semi-infinite time with the same mean squared amplitudes (Figure 10.6 right).

10.4.2 Random sources distributed on a closed curve

The second source configuration is random sources on an arbitrary curve enclosing stations. Now, we observe the potential ψ at $\mathbf{r}_o$ within the circle with radius r (Figure 10.6). Based on the representation theorem, the wave excited by distributed sources outside the circle can be

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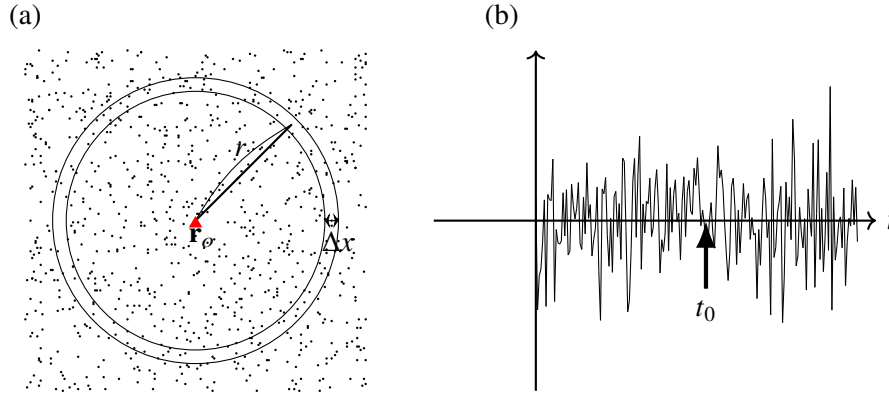


Figure 10.6: Schematic figure for random sources in 2-dimension. (a) Source distribution in space. The red triangle shows the observed station. (b) Potential $\psi(\mathbf{r}_o, t)$ observed at the origin $\mathbf{r}$ at time t . The wave packet at time t_0 is traveled from the source within the concentric circle with radius r shown in (a).

completely reproduced from the stresses and displacements on the circle (e.g.,⁽¹⁾). The contribution of the sources within the circle for the CCF can be neglected because the contribution from the sources outside the circle becomes infinitely large when considering infinitely long times. These features mean that it is equivalent to persistent sources distributed only on the circle. Suppose that uncorrelated random sources with white spectra on an arbitrary surface enclose stations in a heterogeneous medium. In this case, the time derivative of the corresponding CCF represents the exact Green's function between a pair of stations.⁽⁴⁶⁾

The web demo of seismic interferometry at <http://www.eri.u-tokyo.ac.jp/knishida/Seismology/wave2Drandom2.html> simulates this condition (Figure 10.7).

10.4.3 Uncorrelated plane wave incidents

The third source configuration is uncorrelated plane wave incidents from various directions. Assuming that the source-station distances are sufficiently longer than the aperture of the seismic array, the assumption leads to a plane-wave approximation.^{(13), (19)} The above discussions for a homogeneous source distribution lead to the identical formulation of CCFs regardless of the source configurations, as shown in the following subsection. When we consider a heterogeneous source distribution, the first source configuration is the most flexible to represent the source, and the second and third are gradually less flexible. The flexibility of the first two source configurations causes a complex dependence of a CCF on the locations of the two stations. In contrast, the inflexibility of the third source configuration causes a simple dependence of the CCF only on the relative location between the stations (only the distance and the azimuth). In many cases, the ocean swell activities are far enough away from the stations to approximate the phenomenon well. For simplicity, we will use the third source configuration to represent CCFs based on plane wave incidents in the

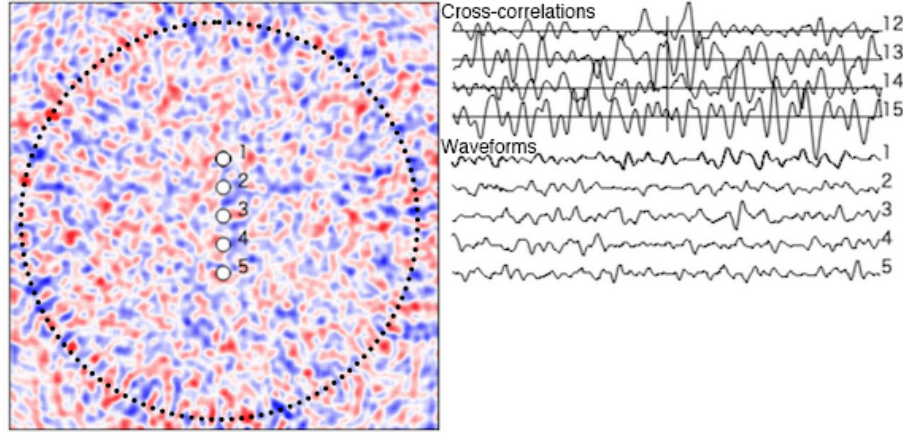


Figure 10.7: Example of a wavefield when random external forces are distributed on a circle. Excitation is assumed to be continuous in time.

following subsections.

Here, for simplicity, we again assume that the time average is the same as the ensemble average. That is, we suppose that an external force $f_i^k(\omega)$ is distributed on the circumference. Let us consider the situation where such excitation of seismic waves is repeated N times, and we take their ensemble average. Assuming that an external force f_i is completely uncorrelated with other excitation sources f_j , and that f is white, we can write

$$\langle f_i^{k*} f_j^k \rangle_k = \delta_{ij} f_0^2, \quad (10.32)$$

where $\langle \cdot \rangle_k$ represents the ensemble average over k . From this, the cross-spectrum (the Fourier transform of the cross-correlation function) Φ_{12} can be written as

$$\begin{aligned} \Phi_{12}(\omega) &\equiv \langle \Phi_{12}^k \rangle_k = \sum_{i,j} G^{2D*}(\mathbf{r}_1 - \mathbf{r}_i, \omega) G^{2D}(\mathbf{r}_2 - \mathbf{r}_j, \omega) \langle f_i^k(\omega) f_j^k(\omega) \rangle_k \\ &= \sum_i G^{2D*}(\mathbf{r}_1 - \mathbf{r}_i, \omega) G^{2D}(\mathbf{r}_2 - \mathbf{r}_i, \omega) f_0^2. \end{aligned} \quad (10.33)$$

If the number of excitation sources is sufficiently large and the sources surround the observation points, the summation in the above equation can be replaced by a line integral:

$$\Phi_{12} = f_0^2 \int_{l_s} G^{2D*}(\mathbf{r}_1 - \mathbf{r}_s, \omega) G^{2D}(\mathbf{r}_2 - \mathbf{r}_s, \omega) dl_s. \quad (10.34)$$

This takes the form of spatially integrating the convolution integral of the Green's functions. This equation is the fundamental equation in seismic interferometry.

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10.4.4 Evaluation of the integral: Relationship between cross-correlation and Green's function

To evaluate this integral, let us consider the source distribution shown in Figure 10.8. The position of the source is represented by the distance r_s and angle ϕ_s from the origin.

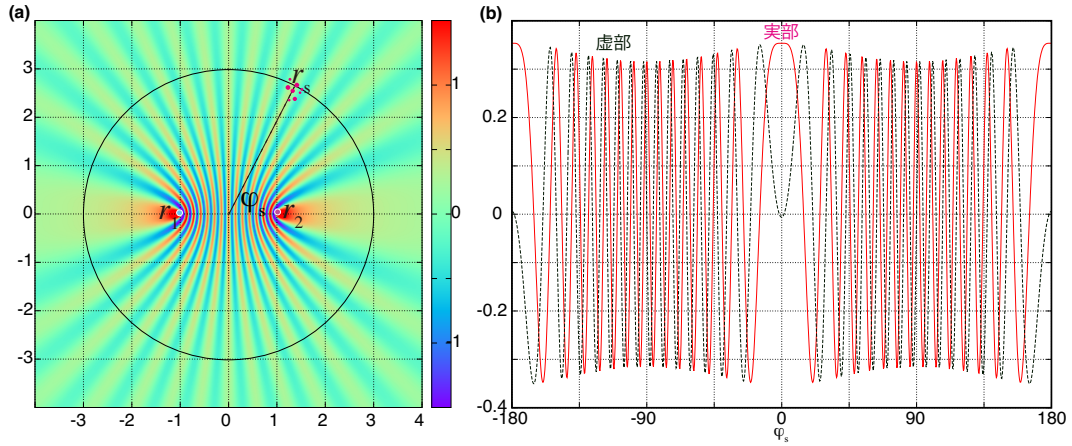


Figure 10.8: (a) Plot of the integrand with respect to ϕ_s at a fixed r_s . (b) Values along the circle at $r_s = 3$ (see Figure 10.8(a)).

For simplicity, we approximate that the Green's function is proportional to $e^{ikr-\pi/4}/\sqrt{kr}$. This approximation is valid when the distance r is longer than the wavelength^{note 10)}. Then,

$$\Phi_{12} \propto \int_{l_s} \frac{e^{ik(r_{2s}-r_{1s})}}{k\sqrt{r_{1s}r_{2s}}} dl_s, \quad (10.35)$$

where r_{1s} and r_{2s} are the distances between station 1 (r_1), station 2 (r_2), and the source r_s , as shown in Figure 10.8. As seen in the equation, the phase difference at stations 1 and 2 for the wave radiated from the point source r_s is determined by the difference in distance divided by the wavelength. The curves of equal phase difference form hyperbolas with the observation points as foci. Thus, along the path passing through the two stations, the phase change is gradual

^{note 10)}Specifically, the Green's function can be written as:

$$G^{2D}(\mathbf{r}, t) = -\frac{1}{2\pi} \frac{H\left(t - \frac{r}{c}\right)}{\sqrt{t^2 - \frac{r^2}{c^2}}}.$$

In the frequency domain:

$$G^{2D}(\mathbf{r}, \omega) = -\frac{i}{4} H_0^{(2)}(\omega r/c) \propto \frac{1}{\sqrt{kr}} e^{ikr-\pi/4},$$

where the wavenumber k is defined as $k \equiv \omega/c$, $H_0^{(2)}$ is the Hankel function of the second kind, and $H()$ is the Heaviside step function.

(stationary point), while in other regions, it oscillates rapidly (Figure 10.8). When considering the problem of random excitation, the contribution from the sources along the path between the two observation points (called the stationary zone) becomes dominant, and the influence of sources in other regions cancels out (method of stationary phase, see e.g., Yomogida, 2007). Moreover, since the imaginary part is antisymmetric with respect to ϕ_s , it cancels out when the source distribution is uniform^{note 11)}.

Below, we try a slightly more intuitive explanation from a different perspective. As shown by the red dots in Figure 10.8, when random sources are distributed in a limited area, the wavefield observed far enough away can be represented by a superposition of multipoles (monopole, dipole, quadrupole, etc.) (known as multipole expansion). Let us consider the case where ϕ_s is around 70° as shown in the figure. In this case, the probability of a node falling between the two stations is random. Therefore, the probability that the waves from r_s are in-phase at the two stations is equal to the probability that they are out-of-phase. On the other hand, when ϕ_s is 0 or π (the case of the stationary zone), since the direction to the two stations is the same from the source's perspective, the waves from r_s are always in-phase. In other words, only when the excitation sources are located in the stationary zone do they contribute to the cross-correlation waveform.

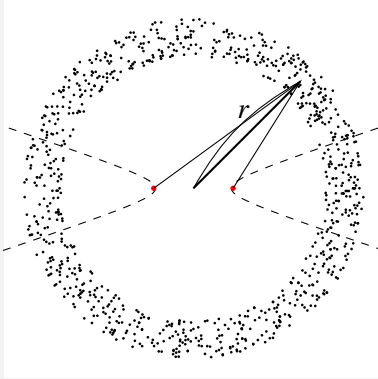
^{note 11)} The rapidly oscillating property of the integrand is important. This is because even if the excitation sources are not uniformly distributed, the contribution from the stationary zone is expected to be dominant. The robustness of the cross-correlation waveform against the source distribution is a crucial point in data analysis.

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Problem 9.2

Although evaluated in the frequency domain above, let us consider it in the time domain to understand the shape of the cross-correlation waveform. Here we consider the situation as shown in the accompanying figure. The term written as $G^{2D*}(\mathbf{r}_1 - \mathbf{r}_s, \omega)G^{2D}(\mathbf{r}_2 - \mathbf{r}_s, \omega)$ in the frequency domain can be represented by a convolution integral of Green's functions in the time domain (by inverse Fourier transform).

1. Calculate the cross-correlation function when an external force is applied at the position shown in the figure.
2. Consider the case where the external forces are distributed on the circumference shown in the figure. Average the cross-correlation function calculated in (1) with respect to the angular direction, and evaluate the cross-correlation function. Note here that the contribution of the external forces inside the hyperbola in the figure is dominant.
3. Discuss the relationship between the time at which the cross-correlation function reaches its peak value and the distance between the observation points.



The contribution of the integral was roughly estimated above. Here, following Wapenaar and Fokkema (2006),⁽⁴⁷⁾ we consider it further by assuming that u satisfies the radiation boundary condition on l_s :

$$\frac{\partial u}{\partial n} = -iku. \quad (10.36)$$

Combining this with Rayleigh's reciprocity theorem, we can relate the cross-correlation function and Green's function as follows^{note 12)}:

$$\Phi_{12} = f_0^2 \int_{l_s} G^{2D*}(\mathbf{r}_1 - \mathbf{r}_s, \omega) G^{2D}(\mathbf{r}_2 - \mathbf{r}_s, \omega) dl_s, \quad (10.37)$$

$$\sim f_0^2 \frac{\rho c}{2i\omega} \left(G^{2D*}(\mathbf{r}_1 - \mathbf{r}_1, \omega) - G^{2D}(\mathbf{r}_1 - \mathbf{r}_2, \omega) \right). \quad (10.38)$$

^{note 12)} See Wapenaar and Fokkema (2006)⁽⁴⁷⁾ for details.

This takes the form where the derivative of the cross-correlation function is proportional to the Green's function^{note 13)}.

An important point to note here is that attenuation is not considered, and because a radiation boundary condition is used, free-boundary surfaces such as surface reflections cannot be taken into account. Considering a realistic setting, it cannot be interpreted as the Green's function unless excitation sources are also distributed underground surrounding the region of interest. Therefore, for the actual cross-correlation function, the amplitude of body waves will be smaller compared to the Green's function (roughly speaking, an earthquake record). The problem of being unable to extract body waves well, especially surface reflections, can be avoided to some extent by using coda waves (e.g., Tonegawa et al., 2009⁽⁴⁰⁾).

Problem 9.3

Run the web demo of seismic interferometry at <http://www.eri.u-tokyo.ac.jp/knishida/Seismology/wave2Drandom2.html> to observe the wave propagation and discuss its correspondence with the theory.

§10.5 Conditions for application of surface wave tomography

Let us consider the difference in ray geometry between ANT and earthquake surface wave tomography. In the case of regional-scale earthquake surface wave tomography with source-receiver paths, the target area is surrounded by earthquakes, which are limited along active tectonic areas (Figure 10.9 (a)). Because most ray paths travel across the area, the travel-time anomalies reflect the seismic structure along lines over the scale. Because the perturbations of the inferred phase/group velocity map strongly depend on the damping of the tomographic inversion, the absolute velocities tend to be ambiguous. Although the damping problem is common in ray theoretical inversion, including ANT, an initial local 1-D model for ANT constructed from multimode dispersion measurements using subarrays (see section ??) can mitigate the ambiguity.

In contrast, in the case of ANT with a dense array, the distribution of the ray path is uniform (Figure 10.9(b)). Above 0.05 Hz, CCFs with longer separation distances (typically longer than 1000 km) become complex. Ray paths with separation distances shorter than 1000 km enable us

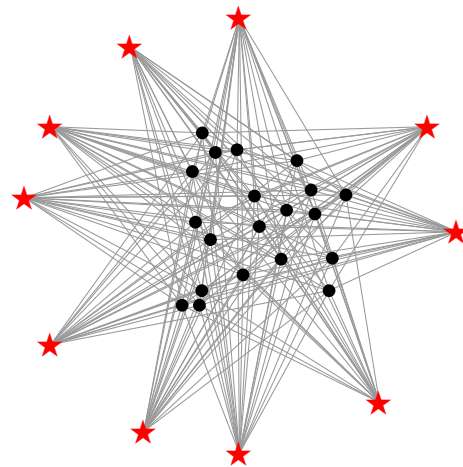
^{note 13)}The spatial autocorrelation method (SPAC method) by Aki (1957) is discussed under the assumption that plane waves are incident randomly from afar. While the general framework is similar to the discussion in this chapter, the details differ. For example, based on the SPAC method for the 2-D case (see Nakahara, 2006), the Hilbert transform of the cross-correlation function yields the Green's function. This originates from the fact that the wave incident from afar differs from the Green's function and the plane wave $\exp(i(\omega t - \mathbf{k} \cdot \mathbf{x}))$.

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to infer tomographic maps even in regions with strong lateral heterogeneities above 0.1 Hz. This situation is similar to earthquake tomography using the two-station method (e.g., [(9), (12),]), which measures the phase differences between seismograms for two stations on a common great-circle path. The drawbacks of the two-station method are (1) that it requires both stations to be located close to the common great circle path, which reduces the available ray paths, and (2) that the longer earthquake-receiver distance tends to cause the wave propagation to enter the regime of multiple scattering at a higher frequency (Figure ??).

The shape of the sensitivity kernel is elliptical along the ray path, and the width at the midpoint is proportional to $\sqrt{\lambda r_{12}}$, where λ is the wavelength. Because the lengths of ray paths of earthquake data become longer in general, the widths become wider than those of ANT. The wider width tends to average the anomalies within the fat ray, and the averaging over a long distance makes restoring the small-scale image difficult.

(a) Earthquake surface wave tomography



(b) ANT

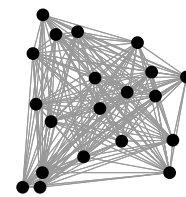


Figure 10.9: (a) A typical example of earthquake-receiver geometry for earthquakes. The gray lines show ray paths between the earthquakes and the stations. Red stars show the earthquakes, and black dots show stations. (b) A typical example of station geometry for ANT. The lines show ray paths between the station pairs.

The advantages of ANT are the homogeneous path distribution and the shorter ray paths. These features enable us to estimate phase/group velocity maps above 0.1 Hz.

Summary: pros and cons of cross-correlation of microseisms

- We can use information from short-distance paths.
- Even at periods shorter than 0.05 Hz, it is less susceptible to scattering and attenuation effects.

- × If there is a bias in the distribution of excitation sources, apparent travel-time anomalies may occur.

Summary: pros and cons of Ambient noise tomography

- The paths do not depend on the earthquake distribution, and there is less bias in the path density.
- In the case of array observations, it is possible to estimate a good initial model (local 1-D structure).
- × If there is a bias in the distribution of excitation sources, apparent travel-time anomalies may occur. This is particularly serious when estimating azimuthal anisotropy or attenuation structure.

ToDo: Relationship between seismic interferometry using a spring-mass model and Brownian motion

§10.6 An application for seismic monitoring

In considering the dynamics of phenomena occurring inside the Earth, it is extremely important to capture temporal changes in velocity structure. This is because volcanic eruptions and earthquakes are expected to cause changes in stress and strain states, accompanied by changes in velocity structure and anisotropy.

When actually trying to determine temporal changes in underground structure, it is ideal to repeat seismic tomography using controlled sources. However, in many cases this is not realistic. On the other hand, when using natural earthquakes, uncertainties caused by the uncertainty of the source and the bias in the source distribution lead to uncertainties in the velocity structure. Therefore, even if a temporal change is apparently observed, it is unclear whether it is just noise or a true velocity change.

In contrast, seismic interferometry is a very effective method for detecting temporal changes in underground structure. This is because by calculating the cross-correlation function over divided periods, we can virtually observe repeating earthquakes. Let us consider the simplest example. We observe earthquakes at two points on the surface, and continuously monitor changes in the cross-correlation function of the records. In this case, localized structural changes between the two points can be monitored moment by moment. Although we will not discuss the details here, pioneering studies include Sens-Schönfelder and Wegler (2006) (Merapi volcano, Indonesia)⁽³²⁾ and Wegler and Sens-Schönfelder (2007) (analysis of the Chuetsu earthquake).⁽⁴⁹⁾

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

§10.7 Practical problems when applying actual data

10.7.1 Azimuthal dependence of incident waves

To consider the azimuthal dependence of the incident wave amplitude, we consider Cox's equation⁽⁷⁾

$$\sum_{m=0}^{\infty} i^m J_m \left(\frac{\omega r}{c(\omega)} \right) [a_m(\omega) \cos(m\zeta) + b_m(\omega) \sin(m\zeta)] \quad (10.39)$$

In this case, the travel-time anomaly can be written as

$$\begin{aligned} \delta t &= \frac{B''(0)}{2t\omega^2 B(0)} \text{ for the causal part} \\ &= \frac{B''(180)}{2t\omega^2 B(180)} \text{ for the acausal part} \end{aligned} \quad (10.40)$$

(Weaver et al., 2009).⁽⁴⁸⁾ Using this equation, it is possible to correct the anisotropy of the incident wave.

10.7.2 Finite frequency effects

When applying seismic interferometry, it is not obvious whether it is sufficient to consider the same sensitivity kernel as for normal earthquakes.⁽⁴⁴⁾ It is known that when the excitation sources are isotropic and homogeneous, the sensitivity kernel of surface waves is the same as that of earthquakes,⁽²⁶⁾ but when the excitation sources are uneven, strict corrections are necessary.

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